

Maths Trail at the Dundee Botanic Gardens

Maths Week Scotland - Dundee University

Welcome! This site was created in 2026 at the University of Dundee for our Maths Trail at the Dundee Botanic Gardens, as part of the wider Maths Week Scotland programme of events. It is designed to be used while exploring the Botanic Gardens and using a smart-phone to scan QR barcodes taking you to individual pages here.

Phyllotaxis

Plants arrange their leaves around a stem in regular patterns. The angle between one leaf and the next is called the divergence angle, and this pattern of leaf arrangement is known as phyllotaxis. Many different plant species share similar phyllotaxis patterns, and these can be described using a small number of mathematical patterns.

Which of the following leaf arrangement pattern do you find in this plant? Please record the number of each leaf arrangement pattern in the table provided on the Maths Trail map: **distichous** (180°), **Fibonacci spiral** (137.5°), **decussate** (90°), or **tricussate** (60°).

Image from the University of Tokyo that shows the classification patterns:

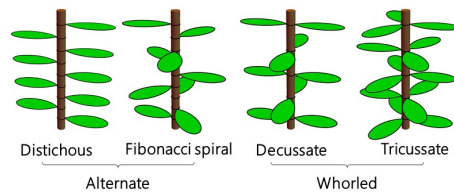


Figure 1: Phyllotactic Patterns

Logarithmic Spiral

Logarithmic spirals in plants are a striking example of geometry in nature, where botanical structures grow according to the Golden Ratio and the Fibonacci sequence.

Spiralling Succulents

Many varieties of succulents arrange their leaves outward from a central point along an expanding logarithmic trajectory to ensure uniform access to sunlight and water collection.

Image from the Fractal Dimensions in Circular and Spiral Phenomena that shows logarithmic spiral in Aloe Polyphylla:



Figure 1: Logarithmic spiral in Aloe Polyphylla

Symmetry in Plants

Links to Group Theory.

Fractals

A fractal is a structure that possessed the property of **self-similarity**. Specifically, it refers to a structure where the exact same shape repeatedly appears when the entire figure is broken down into several smaller parts.

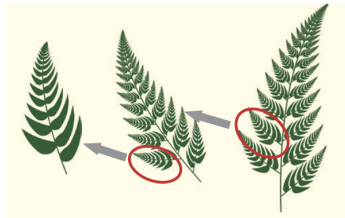


Figure 1: Fractal pattern in fern leaf

While fractal patterns and shapes appear complex at first glance, they are actually the result of repeating simple rules and operations.

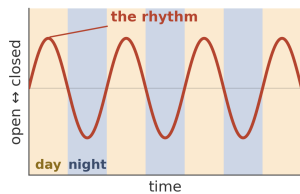
While fractals are a concept rooted in geometry, the fractal approach can also be applied to analyse various phenomena, such as stock prices, currency fluctuations, and social structures.

Many examples of fractals can also be observed in the natural world. These include a wide variety of examples, such as **snowflakes**, *seashells* (like turban shells and nautilus), the shapes of cumulonimbus clouds, and the circulatory system in the human body (including the heart, blood vessels, and lymphatic vessels).

Circadian Rhythm

If you were to watch the water lily here through a full day you would see its flowers open in the morning and close again in the evening.

However, if it was left in a room with the lights left on permanently, it would keep opening and closing on roughly the same rhythm for a while, gradually drifting out of step with real time. The water lily isn't just reacting to light like a switch, it has its own internal molecular clock that is entrained to day-night cycle.



Your cells also have an internal molecular clock. And if you've ever experienced jet lag, your internal molecular clock has become out-of-sync with your new day-night cycle. It takes a number of days of light stimulus in your new environment before your body clock syncs with its environment.

Understanding how oscillators can become entrained becomes mathematical problem and usually depends on the type of oscillator and it how it responds to the entraining stimulus.

Fibonacci Numbers

Look at a pinecone from the bottom. How many spirals go in the clockwise direction (green lines in the photo)? How many spirals go in the counter-clockwise direction (yellow lines)? Wouldn't you expect that they would be the same?

Image from the MENSA for Kids that shows Fibonacci numbers in pinecone:

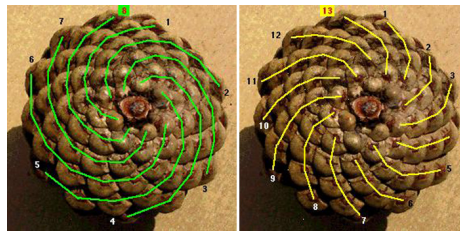


Figure 1: Fibonacci Numbers in Pinecone. There are 8 spirals in the clockwise direction and 13 spirals in the counter-clockwise direction.

The number of spirals found on pine cones are almost always Fibonacci numbers, which are

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...

Bird Song

Listen to the birds around you on the trail. Each song is a pressure wave travelling through the air, picked up by your ear (or a microphone) as a single wiggling signal that changes moment to moment. Buried inside that one signal is a mixture of many pure tones at different frequencies, all layered on top of one another – a blackbird’s song is a fast, fluid run through dozens of them.

Turning a sound into a picture

A short window of any recorded sound can be decomposed into a sum of sine waves of different frequencies. The *Fourier transform* is the tool that performs this decomposition: feed it a chunk of samples, and it hands back how much of each frequency is present.

Do this over and over on a sliding window as the recording plays out, and stack the results side by side, and you get a **spectrogram** – a picture with time along one axis, frequency along the other, and colour showing how strong each frequency is at each moment. A pure whistled note shows up as a single bright horizontal band. A complex trill, with several frequencies sliding around at once, shows up as a tangle of curved streaks. Birdsong is a particularly rich example: many species sing well above the pitch of human speech, and pack in rapid frequency sweeps that are hard to pick out by ear but leap out immediately in a spectrogram.

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The Analemma Project

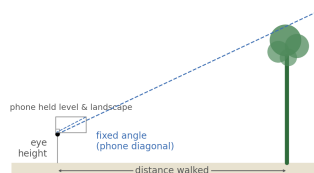
Artist Lucy Smith created the Analemma Project sculpture. She completed her PhD “Making the Conceptual Tangible: The Role of Art in Communicating Science” here at the University of Dundee.

The Analemma is an unequal figure-of-eight shape, created by tracking the sun’s position in the sky at the same time each day over a period of one year.

Why doesn’t the Earth fall onto the Sun?

The Analemma traces that figure-of-eight because the Earth’s orbit isn’t a perfect circle travelled at constant speed – it’s an ellipse, and Earth moves faster when it’s closer to the Sun. Working out exactly *why* orbits look like this took the better part of three centuries. **Galileo** discovered that not everything orbits the Earth; **Kepler**, worked out the precise shape of orbit based on observations; **Newton** explained *why* from a single law of gravity; and **Einstein**’s’ general relativity is needed to explain a tiny mismatch between Newton’s theory and Mercury’s orbit.

Estimating the Height of a Tree using Trigonometry



Suppose you need to measure the height of the tree in front of you, standing on the ground with nothing but your own two feet and the phone that you are using on this tour.

The idea

Everything below comes down to one right-angled triangle:

- your eye,
- the point on the tree that is at the same height as your eye,
- the top of the tree.

Given this triangle, the vertical height of the tree can be estimated given the walking distance to the tree and the slope of the hypotenuse, i.e.

$$\text{tree height} = \text{eye height} + \text{distance} \times \tan(\text{angle})$$

Why leaves crinkle: differential growth in a disc



Why are some leaves more wrinkled than others? And what causes this to happen?

Consider a circular disc of radius, R .

The perimeter is $2\pi R$ whilst the area of the circle is πR^2 . It can be shown that this circle is the most efficient way to contain the area πR^2 .

Now imagine distorting the disc so that the circumference is wrinkled. The circumference of the wrinkled shape is larger than $2\pi R$.

In the plant at this stop, notice that the leaves are wrinkled rather than smooth. Why does evolution find this shape for this particular plant and how does it happen?

One way to make a wrinkled surface is to have different regions of the surface grow at different rates. Essentially, take a circle and allow the outer region to grow at a faster rate than the inner region. The effect of the differential growth is that the faster growing region wrinkles.

The Travelling Salesman problem

Between the Fibonacci stop and this one, the trail passes four other stops – Phyllotaxis, Analemma, Königsberg and Curvature. Suppose you wanted to visit each of the stops once whilst minimising the distance that you travel.

This is the **Travelling Salesman Problem** (TSP) – one of the most famous problems in mathematics and computer science. It sounds simple: just try every order of stops and keep the shortest. With 5 stops there are only 12 distinct routes to check. With 10 stops there are already 181,440. With 20 stops there are over 60 quadrillion – so many that a computer checking a billion routes every single second would still need about two years to get through them all. Trying every option, which works fine for a handful of stops, becomes hopeless almost immediately – and no computer, however fast, escapes that by brute force alone.

Real versions of this problem sit behind delivery route planning, circuit board drilling, and DNA sequencing.

Helicopter Seeds

The seeds from maple, elm and ash trees have curved wings to let them stay in the air longer. The wind will pick up these seeds and carry them to suitable ground. These seeds twirl and spin around. They will then drop and sprout a new tree. Some people call these seeds **helicopter seeds** because they look like and can spin like a propeller.

Here you will learn how to make your own spinner.

1. Fold the middle flap in the opposite direction of the other two flaps. Put the paper clip on the end of the middle flap.
2. Find a maple seed spinner. Drop both spinners from the same height and compare how they spin. Does your spinner work as well as nature's?

Seven Bridges of Königsberg

In the 1700s the city of Königsberg was built around two branches of a river, with an island sitting where they met. Seven bridges connected the four separate pieces of land – and the people of the city had a favourite Sunday puzzle: could you walk through the city and cross every one of the seven bridges exactly once?

Nobody could ever do it. Nobody could ever explain *why*, either – until a mathematician called Leonhard Euler looked at the problem in 1736, and realised something surprising: it doesn't matter how big the islands are, how long the bridges are, or what order you visit things in. All that matters is *which* pieces of land are joined to *which*, and by how many bridges. That single idea – throw away the shape and the distance, keep only the connections – was the birth of an entire branch of mathematics: graph theory, and later, topology.

End of Maths Trail Questionnaire

Display at the end of Maths Trail (near the reception?). Links to Microsoft form, which asks feedback and also asks to enter the counted numbers for Phyllotaxis patterns.

Share your trail on Instagram

Snap a photo of your favourite discovery from today's trail and share it on Instagram. If you tag **@dundeemaths** then some of our undergrads will be able to answer any questions that you might have about the tour.

If you tag @dundeuni you can let the University of Dundee know how much appreciate Mathematics and the Botanic Gardens!